

Report

of

Development of AI-Powered Automation Solutions for KDI Power

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE (AI/ML)



By

Roll Number of Student: **CS-2341412**

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Batch: **2023–27**

Organisation of Internship:

KDI Power Private Limited

New Delhi, India

1 Candidate's Declaration

I, **Girish Kumar Yadav**, hereby declare that the internship report titled “**Development of AI-Powered Automation Solutions for KDI Power Private Limited**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering from IILM University, Greater Noida.

The work presented in this report is based on the technical work undertaken by me during my internship at KDI Power Private Limited in the domain of Artificial Intelligence and Automation. All references and external resources used in preparing this report have been appropriately acknowledged. This report has not been submitted to any other institute for the fulfilment of any degree or diploma requirement.

Signature

Student Name: Girish Kumar Yadav

Roll No.: CS-2341412

2 Acknowledgement

I would like to express my sincere gratitude to everyone who supported and guided me throughout my internship at KDI Power Private Limited. This internship provided me with an opportunity to apply academic knowledge of Artificial Intelligence, Machine Learning, backend development and automation to practical industrial problems.

I am especially thankful to **Mr. Vijay Gautam, Administrative Manager, KDI Power Private Limited**, for his guidance, support and feedback throughout the internship. His understanding of the organisation's operational requirements helped me identify practical areas where software, automation and artificial intelligence could provide meaningful value.

I am also grateful to the faculty members of the School of Computer Science and Engineering, IILM University, for providing the academic foundation and encouragement necessary to undertake this internship. The faculty supervisor is listed as **TBD** pending final confirmation.

I would also like to thank the employees and teams at KDI Power who provided operational context and feedback during development and testing. Finally, I express my gratitude to my parents, family and friends for their continuous encouragement and support.

Signature

Student Name: Girish Kumar Yadav

Roll No.: CS-2341412

3 Internship Completion Certificate

The official internship completion certificate issued by **KDI Power Private Limited** is attached with the final printed copy of this report.

The certificate bears certificate number **KDIP/INT/2026/AI-014**, was issued on **2 September 2026**, and was authorised by **Mr. Vijay Gautam, Administrative Manager, KDI Power Private Limited**. The certificate recognises work associated with the KDI Power WhatsApp Business Messaging System, PDF Editor and LinkedIn Lead Generator.



KDI POWER
wires and cables
ISO 9001, 14001, 45001 Certified
MFG: ALL TYPES OF IT CABLES

TO WHOM IT MAY CONCERN

This is to certify that Mr. GIRISH KUMAR YADAV has successfully completed a two-month internship in **Artificial Intelligence & Automation** with **KDI Power Private Limited**, from **29 June 2026** to **31 Aug 2026**. During this period, he demonstrated strong technical aptitude, initiative, and commitment, and independently designed, developed, and deployed the **KDI Power WhatsApp Business Messaging System, Pdf Editor & LinkedIn Lead Generator**.

KEY RESPONSIBILITIES & CONTRIBUTIONS

- **Automation Development:** Designed and deployed an end-to-end WhatsApp Business messaging workflow on the Meta Cloud API, covering template submission, approval tracking, and campaign dispatch.
- **Backend & Database:** Modelled and maintained the Supabase schema for customers, templates, campaigns, and message logs, including authentication and row-level access rules.
- **Integration Engineering:** Implemented webhook handlers for inbound messages and delivery callbacks, enabling automated replies to product enquiries and quotation requests.
- **Testing & Deployment:** Carried out functional testing, error handling, and live deployment of the system for internal business use with minimal supervision.
- **Documentation & Handover:** Prepared setup documentation and a usage walkthrough for the internal team.

INTERNSHIP EVALUATION

Parameter	Observation	Performance
Technical Skills	Strong grasp of API integration, database design and automation logic.	Satisfactory
Work Quality	Delivered a working, deployed system within the internship tenure.	Satisfactory
Initiative	Worked independently and proposed practical workflow improvements.	Satisfactory
Communication	Explained technical decisions clearly to non-technical stakeholders.	Satisfactory
Professional Conduct	Displayed discipline, punctuality and adaptability throughout.	Satisfactory

Overall, Mr. Girish Kumar Yadav has successfully met the objectives of the internship programme. We place on record our appreciation for his contribution and wish him continued success in all his future endeavours.

Date of Issue: 2nd Sep 2026
Place: Delhi
Certificate No.: KDIP/INT/2026/AI-014



VIJAY GAUTAM
Administrative manager, KDI Power Pvt Ltd

KDI POWER PRIVATE LIMITED

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4 Project Description

4.1 Introduction

Artificial Intelligence and automation have increasingly become important technologies for organisations seeking to improve operational efficiency, reduce repetitive manual work and make business processes more responsive. In an industrial organisation, AI can combine large language models, retrieval-augmented generation, browser automation, document intelligence, conversational interfaces and workflow automation to address practical business requirements.

KDI Power Private Limited operates in the electrical wire and cable manufacturing sector. Its product activities include LT Power Cables, XLPE cables, Aerial Bunched Cables, Control Cables, House Wires and Rubber Cables. The organisation has business processes spanning customer enquiries, sales lead generation, competitor monitoring, technical documentation and internal operational coordination.

During my internship, I worked as an **AI Operations Intern** in the domain of **Artificial Intelligence and Automation**. The internship was conducted from **29 June 2026 to 31 August 2026** and followed a hybrid working model. The working schedule was Monday–Saturday from 10:00 AM to 6:00 PM.

The central objective was to identify repetitive and information-intensive business processes and convert them into automated, intelligent software systems. The internship resulted in five interconnected automation solutions covering customer communication, B2B lead generation, competitive intelligence, technical document processing and internal task management.

The five major systems were:

1. KDI Power AI WhatsApp Assistant and Sales Dashboard.
2. KDI Lead Intelligence and B2B Lead Finder.
3. KDI LinkedIn Tender and Government Scheme Tracker.
4. AI-Powered PDF Reconstructor and Editor.
5. Google Sheets and WhatsApp Task Automation Engine.

4.2 Organization Profile

KDI Power Private Limited is an electrical wire and cable manufacturing organisation based in New Delhi. Its product range includes LT Power Cables, XLPE cables, Aerial Bunched Cables, Control Cables, House Wires and Rubber Cables.

The operational nature of a manufacturing and cable business creates multiple information flows. Customers require product information and quotations, sales teams

require qualified leads, business-development teams need competitor and tender visibility, technical teams work with product datasheets, and factory teams require timely task coordination.

The internship was positioned within the organisation's **Artificial Intelligence and Automation** activities. The industry mentor was **Mr. Vijay Gautam, Administrative Manager**. The faculty supervisor is currently **TBD**.

The internship followed a **hybrid working model**, with working hours from **10:00 AM to 6:00 PM, Monday to Saturday**.

4.3 Internship Objective

The primary objective of my internship at KDI Power Private Limited was to apply my knowledge of Artificial Intelligence, Machine Learning, backend development and automation to solve practical business and operational problems.

The internship objectives were:

1. Identify repetitive business processes suitable for automation.
2. Develop AI-powered systems capable of understanding customer messages, technical documents and business information.
3. Automate customer enquiry handling and structured sales lead capture through WhatsApp.
4. Build an automated B2B lead-discovery pipeline.
5. Automate competitive intelligence and industry monitoring.
6. Develop intelligent document-processing capabilities for scanned technical PDFs.
7. Automate internal task reminders and completion tracking.
8. Gain practical experience in testing, deployment and maintenance of AI-enabled applications.

4.4 Problem Statement

Before automation, several business activities required substantial manual effort. Customer enquiries received through messaging channels could require direct human responses, while lead discovery could involve manually searching online sources and collecting company information into spreadsheets.

Competitive monitoring similarly required employees to inspect competitor and industry posts manually. Technical documents such as scanned cable datasheets could not always be conveniently edited using conventional PDF editors. Internal operational tasks required follow-up communication between managers and employees, creating a need for better acknowledgement and completion tracking.

These challenges were addressed through five major systems covering customer communication, B2B lead discovery, competitive intelligence, technical documents and operational task management.

4.5 Objectives of the Five Projects

1. Develop an AI-powered WhatsApp assistant for customer support and lead capture.
2. Build a B2B lead intelligence pipeline for distributors, utilities and EPC contractors.
3. Automate LinkedIn monitoring for competitors, tenders and government schemes.
4. Develop an AI-powered PDF reconstruction and editing system.
5. Connect Google Sheets and WhatsApp to automate internal task reminders and status tracking.

4.6 Scope of the Internship

The scope covered system design, backend development, AI integration, prompt engineering, RAG, databases, browser automation, web scraping, document processing, frontend development, cloud deployment, scheduled workflows, testing and handover.

- Conversational AI and WhatsApp automation.
- Lead extraction and qualification.
- Google Maps and website data collection.
- LinkedIn browser automation.
- Tender and government-scheme intelligence.
- OCR and document layout processing.
- Vision-language-model integration.
- Editable document reconstruction.
- Google Sheets task management.
- Cloud deployment and scheduled automation.

4.7 Technologies and Tools Used

Table 4.1. Major technologies and tools used during the internship.

Layer	Technology	Role
Programming	Python 3.10+	Core applications and automation.
Backend	FastAPI	APIs, webhooks and backend services.
Frontend	React 18	Interactive PDF editor.
Frontend	HTML/CSS/JavaScript	Sales dashboard and interfaces.
Database	Supabase PostgreSQL	Cloud lead, chat and product data.
Vector Search	pgvector	Retrieval of product/catalogue data.
AI	Groq / Llama	Conversational AI and qualification.
AI	Mistral AI	LinkedIn summarisation.
Vision AI	NVIDIA NIM Llama 3.2 90B Vision	Tables and formulas.
Document AI	MinerU	OCR and layout extraction.
Automation	Playwright	Google Maps and LinkedIn automation.
Messaging	Meta WhatsApp Cloud API	Customer and employee messaging.
Cloud	Render	Application deployment.
Automation	GitHub Actions	Scheduled jobs and keep-alive.
Sheets	Google Sheets API v4	Task data management.
PDF	PyMuPDF	PDF analysis and processing.
Rendering	MathJax + Chromium	Mathematical and PDF rendering.
Excel	openpyxl	Structured Excel output.

4.8 System Architecture

The internship consisted of multiple systems rather than one monolithic application. The systems share a common architecture pattern: business requirements were translated into specialised automation applications, which used AI models, APIs and data stores to produce structured business outputs.

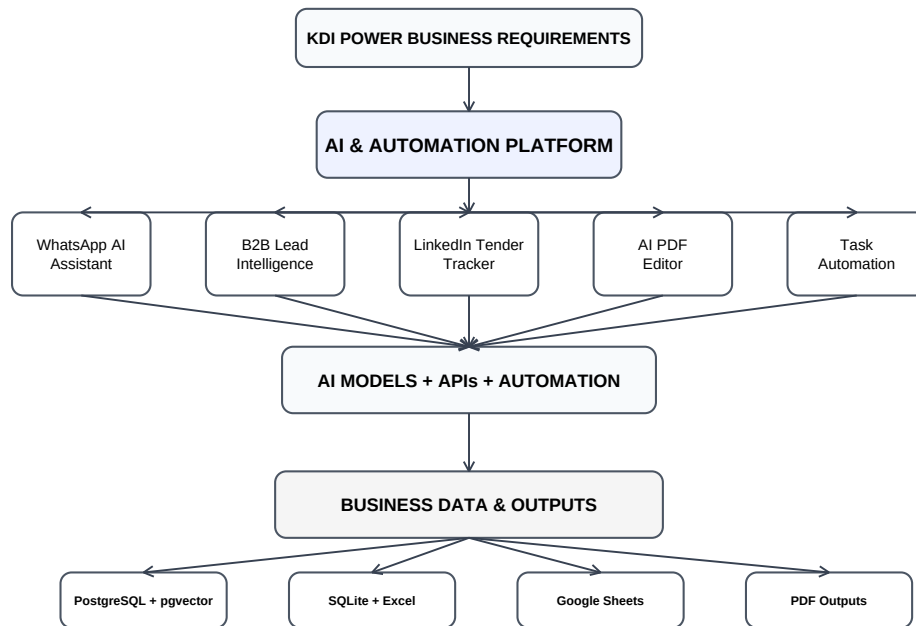


Figure 4.1. Overall architecture of the AI and automation solutions developed during the internship.

4.9 Project 1: KDI Power AI WhatsApp Assistant and Sales Dashboard

The first project was an AI-powered WhatsApp assistant integrated with a sales dashboard. It was designed to automate customer communication, answer product-related questions and capture structured sales leads.

The application used the Meta WhatsApp Cloud API, FastAPI, Groq models and Supabase PostgreSQL.

4.9.1 Architecture and Workflow

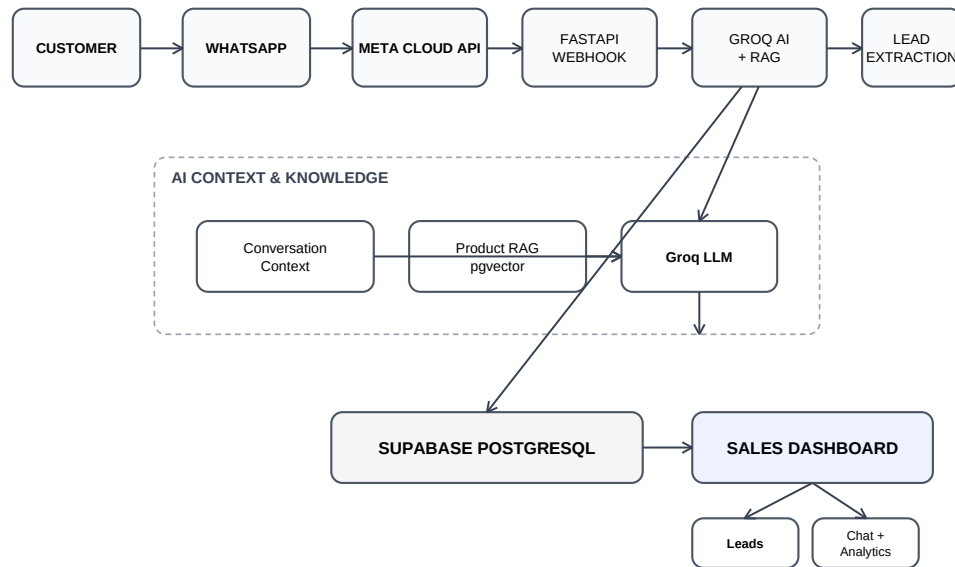


Figure 4.2. End-to-end architecture of the KDI Power AI WhatsApp Assistant and Sales Dashboard.

When a customer sends a WhatsApp message, Meta forwards the message to the FastAPI webhook. The application records the incoming message, retrieves relevant product information, processes the conversation with the AI model and sends an appropriate response.

Lead information is extracted from the conversation and stored in Supabase. The dashboard reads structured lead and chat information and presents it to the sales team.

4.9.2 Artificial Intelligence and RAG

The chatbot used Groq-hosted language models for fast conversational processing. Retrieval-augmented generation was implemented using product/catalogue embeddings stored in Supabase pgvector.

The product data included product name, category, conductor, core count, cable size, indicative price and stock information.

Prompt engineering was used to define the KDI Power persona, catalogue behaviour, lead extraction format and out-of-scope guardrails.

4.9.3 Lead Extraction

Structured lead information included:

Table 4.2. Structured lead information captured by the WhatsApp system.

Field	Description
Name	Customer or contact name.
Phone	WhatsApp contact number.
Firm	Company or organisation.
Email	Email address when available.
Product Interest	Cable or product requested.
Quantity	Required quantity.
Location	Customer location.
Status	New, Contacted, Quoted, Won or Lost.
Timestamp	Record creation/update time.

The chatbot used structured tags such as [LEAD_SUBMIT] and [LEAD_PARTIAL] to identify lead information.

4.9.4 Sales Dashboard

A glassmorphic sales dashboard was developed using HTML5, CSS, Vanilla JavaScript and Chart.js.

It provided a lead inbox, chat transcripts, status filtering, notes, catalogue price and stock editing, inquiry-status charts, daily lead-ingestion charts and product-demand visualisation.

4.9.5 Deployment and Challenges

The application was deployed on Render with Supabase providing cloud database functionality. The WhatsApp webhook was configured with the Meta Cloud API.

The WhatsApp 24-hour service window required approved templates for messages outside the permitted customer-service window. Render cold starts were addressed with keep-alive automation.

4.10 Project 2: KDI Lead Intelligence and B2B Lead Finder

The second project was a B2B lead discovery and qualification pipeline designed to support expansion of KDI Power's business network.

The system targeted cable distributors, electrical wholesalers, utilities and EPC contractors in selected markets in India and South Africa.

4.10.1 Three-Tier Architecture

The system followed three major stages: Discovery, Intelligence and Persistence.

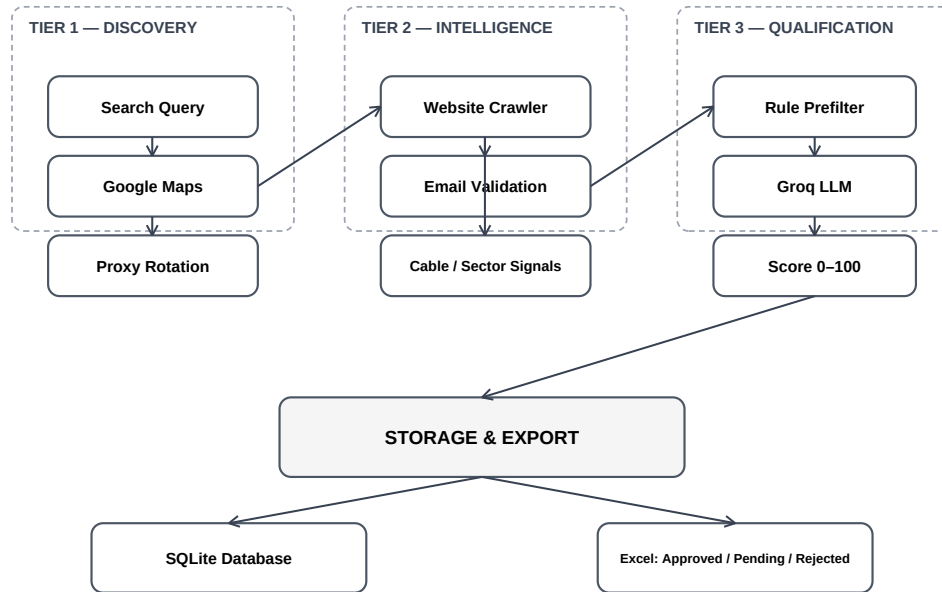


Figure 4.3. Three-tier architecture of the KDI Lead Intelligence and B2B Lead Finder.

4.10.2 Geographic Scope

Indian target locations included Mumbai, Delhi, Bangalore, Chennai and Pune. South African target locations included Johannesburg, Cape Town, Durban, Pretoria and Port Elizabeth.

4.10.3 Lead Discovery and Website Intelligence

Playwright with Chromium was used to automate Google Maps searches. The system extracted company names, websites, phones, emails, addresses, cities, countries, ratings and reviews.

Requests and BeautifulSoup were then used for website intelligence. The pipeline extracted cable-related keywords, product references and contact information, followed by email validation.

4.10.4 AI Qualification and Scoring

A two-stage qualification process was implemented. A rule-based prefilter identified useful signals and an LLM classified companies as Distributor, Utility, EPC, OEM or Irrelevant.

A KDI-specific scoring engine ranked leads on a 0–100 scale.

Table 4.3. Lead qualification thresholds.

Score	Category	Interpretation
75–100	Approved	Strong potential lead.
50–74	Pending	Requires additional review.
Below 50	Rejected	Low relevance or unsuitable.

4.10.5 Challenges and Outcome

Google Maps could return blank results or blocking responses after repeated automated queries. Proxy rotation was therefore supported. Website timeouts, dynamic structures and duplicate companies were also handled.

The final output was organised into Approved, Pending, Rejected and Summary Excel sheets.

4.11 Project 3: KDI LinkedIn Tender and Government Scheme Tracker

The third project focused on competitive intelligence and monitoring of market opportunities in the Indian power and cable industry.

4.11.1 Workflow

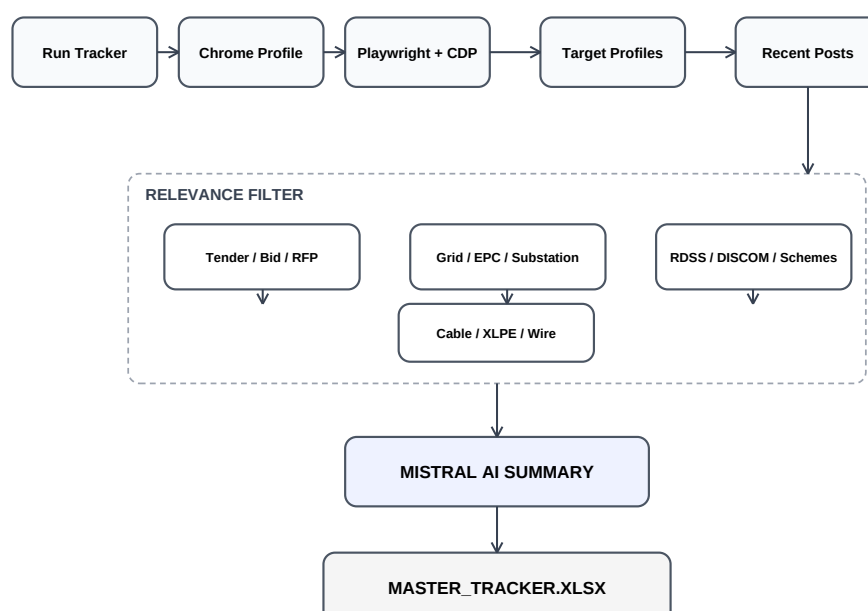


Figure 4.4. Workflow of the LinkedIn Tender and Government Scheme Tracker.

The tracker used Playwright with Chrome DevTools Protocol and a dedicated chrome-data profile to preserve login sessions.

Relevant posts were identified using tender, contract, procurement, electrification, cable and government-scheme keywords. Recruitment terms such as hiring, vacancy and job opening were excluded.

4.11.2 AI Summarisation

Mistral AI was used to generate concise three-to-five bullet summaries. The summaries focused on competitor or author, tender or project, location, cable requirement, scheme and commercial significance.

Exponential backoff was implemented for rate-limit responses.

4.11.3 Excel Intelligence Tracker

The final intelligence was stored in `Master_Tracker.xlsx`. The tracker included collection date, post date, author, post text, post URL, category, summary, status and remarks.

Manual monitoring that could take approximately 2–3 hours daily was reduced to under five minutes for the automated collection and review workflow.

4.12 Project 4: AI-Powered PDF Reconstructor and Editor

The fourth project, **PDFEdit**, addressed editing of scanned technical documents such as cable datasheets, product specifications and procurement documents.

4.12.1 Hybrid Architecture

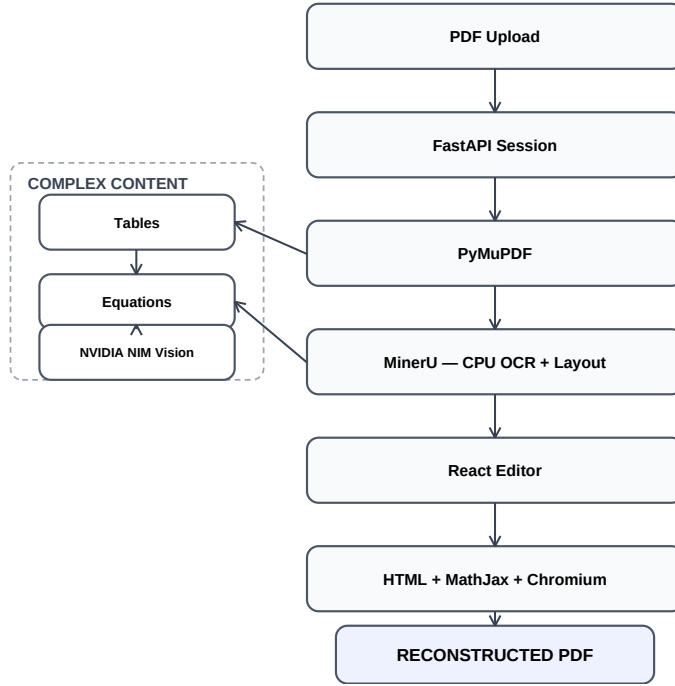


Figure 4.5. Hybrid architecture of the AI-powered PDF reconstruction and editing pipeline.

MinerU was used for OCR and document layout analysis. Complex tables and mathematical expressions were processed using NVIDIA NIM Llama 3.2 90B Vision. The resulting Markdown and LaTeX representations were loaded into a React editor.

4.12.2 Interactive Editing

The React 18 and Vite interface allowed editing of paragraphs, headings, tables and LaTeX expressions. Users could change font family, font size, line spacing, bold, italic and colour, with undo and redo support.

4.12.3 PDF Reconstruction

The reconstructed content followed:

PDF → OCR/Layout → Structured Blocks → Editing → HTML/MathJax → PDF

MathJax rendered mathematical expressions as SVG and Playwright with headless Chromium generated the final PDF.

A major challenge was CUDA memory usage on systems with limited GPU memory. Layout processing was therefore forced onto the CPU while heavy vision processing could be performed through the cloud.

For the demonstrated technical-document workflow, a specification update that could require approximately 45–60 minutes was reduced to under five minutes.

4.13 Project 5: Google Sheets and WhatsApp Task Automation Engine

The fifth project automated internal task assignment, reminders, acknowledgement and completion tracking.

4.13.1 WhatsApp Workflow

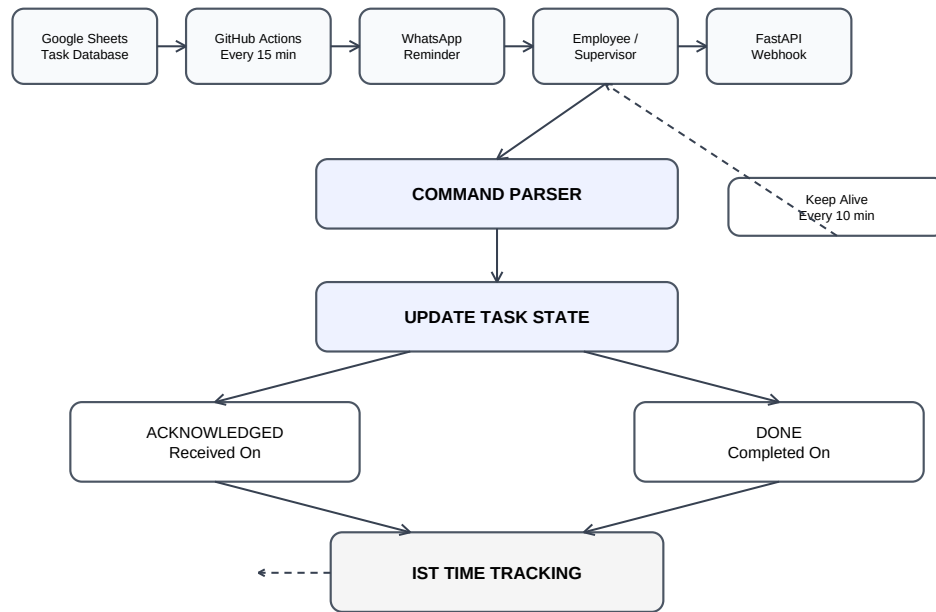


Figure 4.6. Google Sheets and WhatsApp task automation workflow.

The system connected Google Sheets API v4 with Meta WhatsApp Cloud API through a FastAPI webhook.

GitHub Actions periodically checked for tasks requiring reminders. Employees received personalised WhatsApp messages and could respond with commands such as ok, T-101ok, done or T-101done. The system also supported a completion response using the latest open task associated with a phone number.

4.13.2 Task State Management

The task state followed:

Open → Acknowledged → Done

The system recorded ReceivedOn, CompletedOn and calculated TimeTaken using timezone-aware Python timedelta calculations.

All operational timestamps were localised to IST.

4.13.3 Scheduled Automation and Deployment

The reminder workflow was configured to run every 15 minutes and the keep-alive workflow every 10 minutes. The FastAPI webhook was deployed on Render.

The WhatsApp 24-hour service-window restriction, UTC/IST differences, Render sleeping and employee replies without task IDs were addressed through templates, explicit timezone handling, keep-alive jobs and latest-open-task lookup.

4.14 Methodology

The internship followed an iterative development methodology.

4.14.1 Requirement Discovery

The first phase involved understanding the current business workflow and identifying repetitive activities. The focus was on identifying inputs, outputs, constraints and opportunities for safe automation.

4.14.2 Architecture and Technology Selection

Technologies were selected according to the problem. FastAPI was used for APIs and webhooks, Playwright for browser automation, Supabase for cloud data, MinerU for document layout processing and Google Sheets API for lightweight task management.

4.14.3 Implementation

Systems were developed incrementally. Backend functionality was implemented first, followed by AI integration, persistence, interfaces and automation workflows.

4.14.4 Testing and Refinement

Testing covered webhook verification, incomplete inputs, API rate limits, Google Maps blocking, LinkedIn dynamic loading, website timeouts, Excel locks, CUDA memory constraints, WhatsApp service windows, Render cold starts and timezone handling.

4.14.5 Deployment and Handover

Cloud applications were deployed using Render and Supabase. Scheduled processes used GitHub Actions or task schedulers. Portable applications included setup and start scripts so that non-technical users could operate them with minimal configuration.

4.15 Individual Contributions

My major contributions included:

1. Designing and implementing FastAPI webhook services.

2. Integrating Meta WhatsApp Cloud API.
3. Implementing conversational AI using Groq models.
4. Designing prompts, guardrails and lead-extraction formats.
5. Integrating Supabase PostgreSQL and pgvector.
6. Developing the sales dashboard.
7. Building the Google Maps and website lead-discovery pipeline.
8. Implementing proxy rotation and lead deduplication.
9. Developing the KDI-specific lead scoring engine.
10. Implementing LLM-based lead qualification.
11. Developing LinkedIn automation using Playwright and CDP.
12. Integrating Mistral AI for intelligence summaries.
13. Implementing Excel storage, formatting and lock detection.
14. Developing the PDF reconstruction pipeline.
15. Integrating MinerU and NVIDIA NIM Vision.
16. Building the React PDF editing interface.
17. Implementing MathJax and Chromium PDF generation.
18. Integrating Google Sheets API with WhatsApp automation.
19. Implementing task-status parsing and timestamp tracking.
20. Configuring GitHub Actions scheduled workflows.
21. Testing, debugging, deployment and documentation.

4.16 Major Technical Challenges

4.16.1 Third-Party API Limitations

External services such as Meta WhatsApp, Groq, Mistral and NVIDIA NIM have usage restrictions and rate limits. Retries, fallbacks, approved templates and error handling were therefore required.

4.16.2 Browser Automation

Google Maps and LinkedIn are dynamic websites that can detect automated access. Proxy rotation, persistent browser profiles, Chrome CDP connections, dynamic waiting and robust selectors were used.

4.16.3 AI Reliability

Language and vision models can produce incomplete or unexpected outputs. Prompt engineering, validation, structured output requirements and fallback logic were important.

4.16.4 Hardware Constraints

The PDF editor demonstrated that AI document processing can require substantial GPU memory. CPU layout processing combined with cloud-based vision processing created a more practical hybrid architecture.

4.16.5 Deployment Constraints

Cloud applications can experience cold starts while scheduled systems depend on accurate execution times. Keep-alive workflows, scheduled actions and explicit timezone handling addressed these constraints.

4.17 Expected Outcomes and Contributions

Table 4.4. Summary of project outcomes.

Project	Primary Function	Contribution
WhatsApp Assistant	Customer communication	AI product assistance, lead extraction and sales analytics.
Lead Intelligence	B2B lead discovery	Automated discovery, qualification and scoring.
LinkedIn Tracker	Competitive intelligence	Automated monitoring and AI summaries.
PDFEdit	Technical document processing	OCR, layout reconstruction, VLM extraction and editable PDF generation.
Task Automation	Internal operations	Reminders, acknowledgement, completion tracking and time measurement.

4.18 Overall Learning

During the internship, I gained practical experience in designing and developing end-to-end AI and automation solutions from scratch. I strengthened my skills in Python,

FastAPI, React, REST APIs, databases, web automation, web scraping, LLM integration, prompt engineering, RAG, document processing and cloud deployment.

I gained practical experience with Groq, Supabase, Meta WhatsApp Cloud API, Google Sheets API, GitHub Actions, Render, Playwright, MinerU, NVIDIA NIM and Mistral AI.

An important learning was that building an AI system involves much more than connecting an application to a language model. A useful production-oriented system requires structured inputs and outputs, database design, error handling, fallback mechanisms, API-limit handling, authentication, deployment and testing.

The internship also improved my ability to understand business requirements and translate them into technical workflows. I learned to consider whether a system would be understandable and useful to the people who would actually operate it.

4.19 Overall Internship Challenge

One of the major challenges was converting real-world business requirements into reliable and practical automated systems.

The projects involved dynamic websites, third-party APIs, WhatsApp messaging restrictions, AI model limitations, complex document layouts, cloud deployment constraints and different data formats.

Handling API rate limits, browser automation and bot detection, CUDA memory limitations, Render cold starts, Excel file locking and timezone differences required continuous debugging and optimisation.

These challenges helped me develop a stronger understanding of production-oriented software development and problem solving.

4.20 Results and Industry Impact

The developed systems addressed five operational areas: customer inbound sales and lead capture, B2B cable-buyer discovery, competitive and tender monitoring, technical document processing, and operational task tracking.

For the LinkedIn tracker, manual monitoring that could take approximately 2–3 hours per day was reduced to under five minutes for the automated collection and review workflow.

For the PDF editing workflow, a demonstrated specification update that could require approximately 45–60 minutes was reduced to under five minutes.

The task automation system reduced repetitive follow-up and introduced timestamp-based tracking for acknowledgement and completion.

4.21 Future Scope

Potential future improvements include:

1. Integration of the WhatsApp lead system with a complete CRM.

2. Automated quotation generation.
3. Sales forecasting using historical lead data.
4. Expansion of B2B lead discovery to additional countries.
5. Refinement of lead scoring using actual sales outcomes.
6. Integration of additional tender and government-scheme sources.
7. Enhanced PDF table editing and layout preservation.
8. Local vision models for restricted-network environments.
9. Integration with additional enterprise task-management tools.
10. Centralised authentication and role-based access.
11. Centralised monitoring and logging for deployed services.

4.22 Conclusion

The internship at KDI Power Private Limited provided an opportunity to apply Artificial Intelligence, Machine Learning and software engineering concepts to real industrial and business requirements.

Five major systems were developed: the KDI Power AI WhatsApp Assistant and Sales Dashboard, the B2B Lead Intelligence Engine, the LinkedIn Tender and Government Scheme Tracker, the AI-Powered PDF Reconstructor and Editor, and the Google Sheets and WhatsApp Task Automation Engine.

Although each project addressed a different operational problem, they shared a common objective: reducing repetitive manual work through software, AI and automation.

The internship provided practical experience in backend development, APIs, databases, browser automation, web scraping, large language models, RAG, OCR, vision-language models, frontend development, cloud deployment and workflow scheduling.

More importantly, the internship demonstrated the importance of connecting technical solutions with actual business requirements. The experience strengthened both my technical foundation and my understanding of practical AI engineering.

4.23 Certificates of Completion and Other Internship Documents

The internship documentation includes the official internship completion certificate and joining documentation issued by KDI Power Private Limited.

- **Organisation:** KDI Power Private Limited.
- **Domain:** Artificial Intelligence and Automation.

- **Certificate Number:** KDIP/INT/2026/AI-014.
- **Issued Date:** 2 September 2026.
- **Industry Mentor / Signatory:** Vijay Gautam, Administrative Manager.
- **Official Role:** AI Operations Intern.
- **Internship Period:** 29 June 2026 – 31 August 2026.

The official certificate and supporting documents are attached to the physical submission of this report.

5 Bibliography

- [1] Meta Platforms Inc., “WhatsApp Cloud API Documentation,” <https://developers.facebook.com/docs/whatsapp/cloud-api>
- [2] FastAPI, “FastAPI Documentation,” <https://fastapi.tiangolo.com>
- [3] Supabase, “Supabase Documentation,” <https://supabase.com/docs>
- [4] PostgreSQL Global Development Group, “PostgreSQL Documentation,” <https://www.postgresql.org/docs/>
- [5] Groq, “Groq API Documentation,” <https://console.groq.com/docs>
- [6] Mistral AI, “Mistral AI Documentation,” <https://docs.mistral.ai>
- [7] Playwright, “Playwright Documentation,” <https://playwright.dev>
- [8] Google, “Google Sheets API Documentation,” <https://developers.google.com/sheets/api>
- [9] GitHub, “GitHub Actions Documentation,” <https://docs.github.com/en/actions>
- [10] Render, “Render Documentation,” <https://render.com/docs>
- [11] React, “React Documentation,” <https://react.dev>
- [12] Vite, “Vite Documentation,” <https://vite.dev>
- [13] PyMuPDF, “PyMuPDF Documentation,” <https://pymupdf.readthedocs.io>
- [14] MinerU, “MinerU Documentation,” <https://github.com/opendatalab/MinerU>
- [15] NVIDIA, “NVIDIA NIM Documentation,” <https://docs.nvidia.com/nim/>
- [16] MathJax, “MathJax Documentation,” <https://www.mathjax.org>
- [17] Python Software Foundation, “Python Documentation,” <https://docs.python.org/3/>
- [18] Beautiful Soup, “Beautiful Soup Documentation,” <https://www.crummy.com/software/BeautifulSoup/>